

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st November 2020

**Subject CP1 – Actuarial Practice
(Paper B)**

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

You work for an actuarial consultancy firm, which provides specialist actuarial advice to life and general insurance companies. You have just been asked to complete two assignments. Note your answer needs to be related to the information provided in the background information and should discuss/outline areas of professionalism/actuarial advice as required. Limited credit will be given to solutions where generic answers have been given that do not refer back to the information provided.

Assignment 1

Background information

A medium sized insurance company XYZ Insurance selling retail health insurance products has been in operations from past 10 years. Every year, the company does a detailed budgeting exercise whereby next year's revenue and losses are forecasted. This year an adverse deviation in actual loss ratio (defined as incurred claims to earned premium) by 10% has been observed compared to budget.

You have been asked to produce answers to the following questions for XYZ insurance.

- Q. 1)** Outline reasons why XYZ Insurance should monitor its experience. (5)
- Q. 2)** What could be the potential reasons of this adverse deviation in loss ratio of XYZ compared to its budget? (12)
- Q. 3)** What could be the impact of this adverse deviation on various stakeholders of XYZ? (7)

XYZ's competitor ABC Health is also a medium sized 10 year old insurer offering similar products but ABC Health has been showing consistently higher profitability compared to XYZ.

- Q. 4)** What could be the potential reasons for the difference in profitability of two insurers offering similar products? (14)

The Government of the country has now mandated that all health and general insurance companies should offer a Standardized Health product. They will have to offer the Standardized product with defined coverages and policy conditions but are allowed to charge their own premium rates.

- Q. 5)** Discuss the impact on insurers of offering a standardized product. (12)
- [50]**

Assignment 2

Background information

A life insurer in a developing country had sold a single premium non-participating endowment life insurance product a few years back and had invested the premiums in high yielding time deposits with one of the largest banks in the country.

There has been a significant fall in the interest rates in the country since then. A recent change in the investment regulation now prohibits the insurer from investing in bank deposits as an asset class and requires it to exit all existing exposures over a 3 year period.

The insurer is currently using the yield on the bank deposits to derive the interest rates to value the liabilities for statutory reporting for this product.

You have been asked to produce answers to the following questions for the life insurance company.

- Q. 6)** Discuss the implications on the insurer of the change in the investment regulation. (12)

As a result of change in the investment regulation, insurer is contemplating investing in corporate bonds with high yields to back the liabilities of the endowment life product.

- Q. 7)** Outline how the risks of investing in corporate bonds could be mitigated. (11)

- Q. 8)** While performing experience analysis for the single premium product, the company observed that the surrender experience has been lower than expected and its mortality experience has been heavier. Discuss reasons for such mortality and surrender experience? (12)

The Chief Risk Officer of the company has recommended that the company should purchase a cyber-insurance policy considering the increasing cyber risks that the company faces.

- Q. 9)** Define cyber risk and outline the perils that could be covered by such a policy. (8)

- Q. 10)** List internal control measures the CRO could put in place for managing cyber risks. (7)
[50]
